DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

CHAPTER 1

INTRODUCTION

In the past decade, Graphics Processing Units (GPUs) have managed to grab the attention of the researchers, owing to their immense performance improvement over Central Processing Units (CPUs). The performance improvement in GPUs is induced by extensively exploiting thread level parallelism in highly parallel applications. However, this upswing is constrained by large power consumption in GPUs [1]. Contrarily, Near Threshold Computing (NTC) has emerged as the promising energy-efficient design point, as opposed to Super Threshold Computing (STC) [2]. At NTC, the supply voltage is marginally higher than the threshold voltage. The reduction in power consumption at NTC is a result of both reduced supply voltage and consequent reduction in operating frequency. Inevitably, GPUs operating at NTC have evolved as an intriguing research topic [3].

Despite the energy efficiency offered by NTC, there are several design challenges in this regime. The biggest bottleneck at NTC comprises of 10X performance degradation, accompanied with extensive process variation (PV) sensitivity [4]. Though performance degradation is tackled by the parallelism offered by GPUs, the spatial expanse of GPUs make them more vulnerable to PV. To support the massive thread level parallelism GPUs employ latency-hiding with the aid of fast context-switching between threads. A huge register file is required to maintain the latency-hiding ability. Therefore the register file becomes a potential soft target for PV [5]. The most prominent manifestation of PV in GPU register files is the variability in the access latency.

This work explores PV sensitivity of register access latency in GPUs. PV can affect the latency either way, exacerbating the access latency difference between the slowest and fastest accessible register file. Further, PV can affect the operating frequency of Compute Units (CUs) thereby inducing significant variability in their performance. Consequently, a PV-agnostic mapping of register files to CUs can remarkably degrade the processor performance.

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

It is therefore essential to dynamically assess the effects of PV on register access latency as well as CU frequency.

The contributions in this work are as follows:

- The factors plaguing the performance of modern GPGPU applications are investigated. In this pursuit, this work defines certain parameters, based on the performance bottlenecks of GPUs.
- The variability in performance is analysed for different GPGPU applications, by varying the the aforementioned parameters.
- This performance analysis propels the need to design an effective mitigation technique, in order to improve the performance of NTC GPU.

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

CHAPTER 2

RELATED WORK

Previous works in the field of energy-efficient GPU design primarily focus on the STC regime. However, recent researches in this area, related to this work, can be broadly classified into two categories: (a) remodelling GPU register file architecture for energy-efficient GPGPUs, and (b) mitigating process variation impact in GPGPUs. In the first category, Jing et al. propose an embedded-DRAM architecture to reduce the ever increasing cost and power overheads of SRAM based refister files [6]. Majeed et al. exploit the inter-access latency of GPGPU register files to propose a tri-modal control unit [7]. Mao et al. propose a nonvolatile racetrack memory to improve the scalability of GPGPUs [8].Jeon et al. present a register file virtualization technique to reduce the physical area as well as power consumption of architected register files [5]. In the second category, Liang et al. propose variable latency register files to mitigate the effects of process variation on GPGPUs [9]. Basu et al. propose a dynamic parallelization scheme to address the performance variance introduced in GPGPUs due to process variation in the NTC regime [3]. For GPUs capable of incorporating spatial multitasking, Aguilera et al. proposes the assignment of variable resources based on the nature of applications [10]. The effects of PV on CPUs have been addressed in several works [11–17]. GPUs and CPU-GPU heterogeneous systems exhibit a marked difference in architecture and operation. With an increased usage of these systems, it is crucial to check the portability of existing techniques and in fact, come up with newer designs, to curb the effects of variability and make them energy-efficient.

However, to the best of my knowledge, this is the first work to explore the effects of process variation at NTC on register access latency in GPGPUs.

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

CHAPTER 3

MOTIVATION

This chapter, shows that the register file is a potential soft target for process variation in a GPU at NTC (Section 3.1). Using a cross layer methodology (Section 3.2), it is observed that any change in access latency, due to PV, can affect the overall performance of a GPU (Section 3.3). Additionally, there is a steady improvement in performance with shorter register access latency.

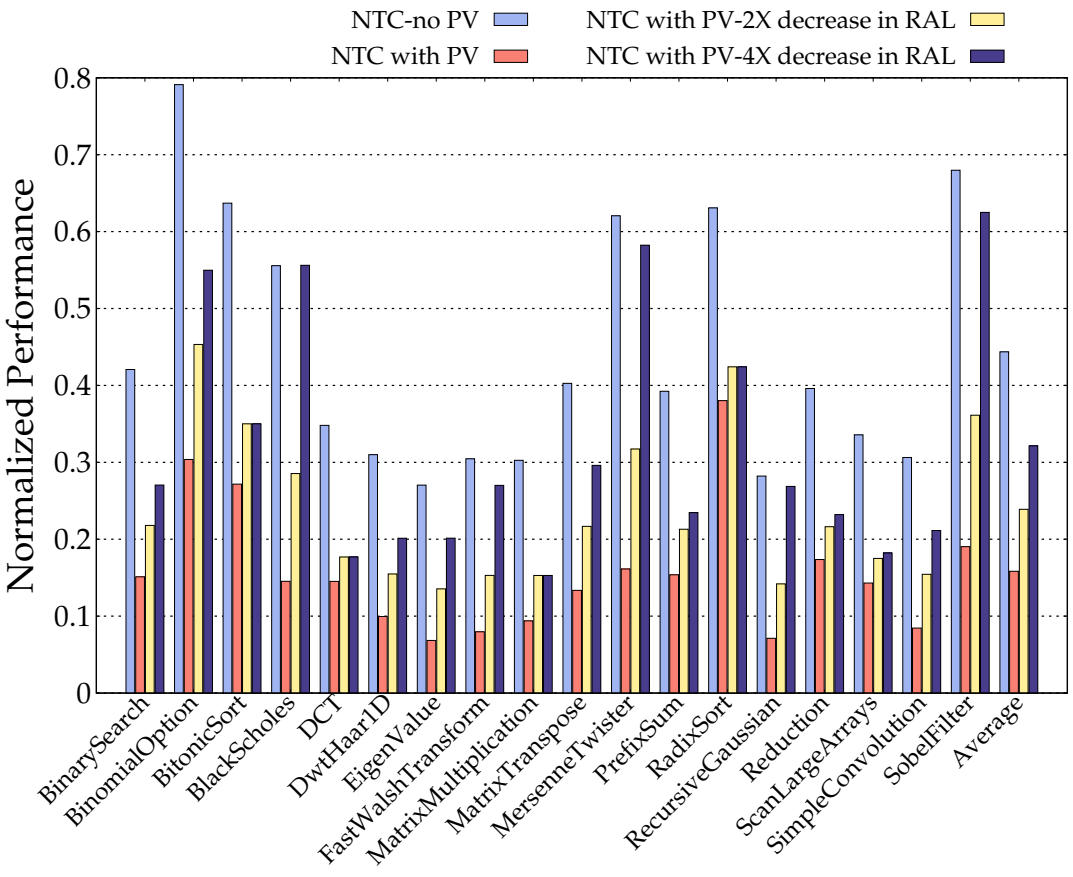


Fig. 3.1: NTC-GPU performance characteristics for : *NTC-no PV; NTC with PV; NTC with PV and 2X improvement in register access latency ; NTC with PV and 4X improvement in register access latency (Normalized to STC GPU).*

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

3.1 NTC-GPU limitations

Individually, GPUs and NTC have emerged as promising design areas. However, their combined potential as a high performance and energy-efficient design paradigm has hardly been explored [3]. NTC operation reduces power consumption but hampers performance due to aggravated PV sensitivity. The following sections address the effects of this variability and its potential in evolving as a performance bottleneck in GPUs.

3.1.1 Process Variation in NTC circuits

The large spatial spread of GPUs makes them inherently vulnerable to PV. The degree and effects of PV on different components of a GPU can be largely different. However, the most prominent manifestation of PV at NTC is in the form of transistor delay variability. The progressive miniaturization of technology nodes further adds to this variability [18]. As a result, PV transpires into a significant performance bottleneck in multicore architectures like GPU.

3.1.2 Register File(RF) Latency in GPU

The size and expanse of RFs in modern GPUs make them prone to PV. The shallow logic depth and large number parallel critical paths of SRAM structures enhances their vulnerability [19]. The SRAM access, read and write times can display marked variation from cell to cell, owing to the effects of PV. The cumulative latency of SRAM read/write access is referred to as register access latency (RAL). This PV induced RAL variability at NTC makes the RFs an intriguing design optimization challenge in NTC-GPUs. Section 3.3 explores how speeding up the overall access latency preserves the performance of a GPU.

3.1.3 Criticality in GPGPU applications

GPUs exploit thread-level parallelism to mask the performance bottlenecks of a single thread. AMD defines a wavefront as the set of threads executing the same instruction with different data. For certain applications, there is an execution time disparity between

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

wavefronts of the same work group[1] and the execution time is dominated by the slowest wavefront. All the wavefronts in a work group need to finish execution before the next work group can be scheduled and the slowest wavefront in the work group imposes a wait time on other wavefronts [20]. Therefore, it becomes an indispensible determinant of overall system performance and for the rest of this work, it is defined as the *critical wavefront*. However, these critical wavefronts do not arise due to lack of synchronization among CUs. The criticality factor primarily springs from varying latencies and pipeline hazards within a work group. Thread criticality is a similar approach for multi-threaded CPU applications to tackle load imbalance on constrained resources, which in turn determines the overall multi-core performance [21]. In the shift from STC to NTC, access latency variation is exacerbated by the increasing parallel bottlenecks at NTC. Hence, RAL would be a controlling factor in determining the critical wavefronts, for a certain timeline[2] in a GPU.

3.2 Methodology

The approach adopted for this work is briefly outlined here. The microarchitectural model of VARIUS-NTV is used for modelling PV in the NTC regime [4]. AMD's Southern Islands architecture based GPU - HD Radeon 7970 model is simulated on Multi2Sim 4.2 [22]. Pre-compiled benchmarks from AMD APP SDK [23] suite is run on this GPU model. The register access latency is 8 cycles for PV-free STC and NTC. Due to the effect of PV the frequency at NTC is reduced upto 75% [24]. Compute Units are scaled upto 4 times at NTC as compared to STC.

3.3 Performance Analysis

Figure 3.1 shows how register access latency can be exploited to improve the performance of the GPGPU applications. The GPGPU application performance is modelled as the inverse of the execution time. All performance values are normalized to a PV-free

[1] Work Group refers to a group of threads/work-items. It is similar to a thread block defined by NVIDIA. Henceforth, these terms are used interchangeably.

[2] GPGPU applications use timelines instead of intervals, as they capture overlap and illustrate critical path.

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

STC GPU. The applications display maximum variation in performance when run on an NTC GPU affected by PV. The cases of 2X and 4X improvement in RAL is considered, for the PV-affected GPU. As observed from figure 3.1, a pronounced performance uplift is observed. The average performance goes up by ∼33% when the register access latency is reduced to 25% of PV-induced value(16 cycles in this work). Reducing the access latency further, by a factor of 8X, shows no appreciable change in performance, as compared to its 4X counterpart. The minimum time taken for access by an SRAM limits this amelioration. With the RAL improvement, *BlackScholes, MersenneTwister* and *SobelFilter* achieve almost comparable performance to an NTC system without PV. PV being a greater concern in NTC than in STC [25], Figure 3.1 clearly shows the scope of register access latency improvement in high performance GPUs operating at NTC. To recuperate this performance loss, it is essential to devise a dynamic paradigm, which can overcome the aforementioned limitations. For further design, this work will be continued to contrive a technique which can give us an improved performance, without jeopardising the power efficiency offered by a GPU operating at NTC.

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

REFERENCES

[1] J. Lucas, S. Lal, M. Andersch, M. Alvarez-Mesa, and B. Juurlink, "How a single chip causes massive power bills gpusimpow: A gpgpu power simulator," in *ISPASS*, April 2013.

[2] R.G.Dreslinski, M.Wieckowski, D. Blaauw, D.Sylvester, and T.Mudge, "Near-threshold computing: Reclaiming moores law through energy efficient integrated circuits," in *Proc. IEEE*, Feb. 2010.

[3] P. Basu, H. Chen, S. Saha, K. Chakraborty, and S. Roy, "Swiftgpu: Fostering energy efficiency in a near-threshold gpu through tactical performance boost," in *Proc. of DAC*, 2016.

[4] U. R. Karpuzcu, K. B. Kolluru, N. S. Kim, and J. Torrellas, "Varius-ntv: A microarchitectural model to capture the increased sensitivity of manycores to process variations at near-threshold voltages," in *DSN*, 2012, pp. 1–11.

[5] H. Jeon, G. S. Ravi, N. S. Kim, and M. Annavaram, "Gpu register file virtualization," in *Proc. of MICRO*, 2015, pp. 420–432.

[6] N. Jing, Y. Shen, Y. Lu, S. Ganapathy, Z. Mao, M. Guo, R. Canal, and X. Liang, "An energy-efficient and scalable edram-based register file architecture for gpgpu," in *ACM SIGARCH Computer Architecture News*, vol. 41, no. 3, 2013, pp. 344–355.

[7] M. Abdel-Majeed and M. Annavaram, "Warped register file: A power efficient register file for gpgpus," in *HPCA*, 2013, pp. 412–423.

[8] M. Mao, W. Wen, Y. Zhang, Y. Chen, and H. H. Li, "Exploration of gpgpu register file architecture using domain-wall-shift-write based racetrack memory," in *Proc. of DAC*, 2014, pp. 1–6.

[9] X. Liang and D. Brooks, "Mitigating the impact of process variations on processor register files and execution units," in *Proc. of MICRO*, 2006, pp. 504–514.

[10] P. Aguilera, J. Lee, A. F. Farahani, K. Morrow, M. J. Schulte, and N. S. Kim, "Process variation-aware workload partitioning algorithms for gpus supporting spatial-multitasking," in *Proc. of DATE*, 2014, pp. 1–6.

[11] R. Teodorescu and J. Torrellas, "Variation-aware application scheduling and power management for chip multiprocessors," in *Proc. of ISCA*, 2008, pp. 363–374.

[12] A. Sasan, K. Amiri, H. Homayoun, A. Eltawil, and F. Kurdahi, "Variation trained drowsy cache (vtd-cache): A history trained variation aware drowsy cache for fine grain voltage scaling," *TVLSI*, vol. 20, pp. 630–642, 2012. [Online]. Available: http://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=5712204

DocuSign Envelope ID: EF8F2518-98B2-4420-8877-069AD3B41EEE

[13] S. Seo, R. G. Dreslinski, M. Woh, Y. Park, C. Chakrabarti, S. A. Mahlke, D. Blaauw, and T. N. Mudge, "Process variation in near-threshold wide simd architectures," in *Proc. of DAC*, 2012, pp. 980–987.

[14] A. Tiwari, S. R. Sarangi, and J. Torrellas, "Recycle: pipeline adaptation to tolerate process variation," in *Proc. of ISCA*, 2007, pp. 323–334.

[15] U. Karpuzcu, N. S. Kim, and J. Torrellas, "Coping with parametric variation at near-threshold voltages," *IEEE Micro*, vol. 33, no. 4, pp. 6–14, July 2013.

[16] J. Torrellas, N. S. Kim, and R. Teodorescu, "Parameter variation at near threshold voltage: The power efficiency versus resilience tradeoff," University of Illinois, Tech. Rep., 2012.

[17] X. Liang, G.-Y. Wei, and D. Brooks, "Revival: A variation-tolerant architecture using voltage interpolation and variable latency," in *Proc. of ISCA*, 2008, pp. 191–202.

[18] L. Chang, D. J. Frank, R. K. Montoye, S. J. Koester, B. L. Ji, P. W. Coteus, R. H. Dennard, and W. Haensch, "Practical strategies for power-efficient computing technologies," *Proc. IEEE*, vol. 98, no. 2, pp. 215–236, 2010.

[19] K. Bowman, S. Duvall, and J. Meindl, "Impact of die-to-die and within-die parameter fluctuations on the maximum clock frequency distribution for gigascale integration," *Solid-State Circuits, IEEE Journal of*, vol. 37, no. 2, pp. 183 –190, feb 2002.

[20] S. Lee and C. Wu, "Caws: criticality-aware warp scheduling for gpgpu workloads," in *PACT*, 2014, pp. 175–186.

[21] K. D. Bois, S. Eyerman, J. B. Sartor, and L. Eeckhout, "Criticality stacks: identifying critical threads in parallel programs using synchronization behavior," in *Proc. of ISCA*, 2013, pp. 511–522.

[22] R. Ubal, B. Jang, P. Mistry, D. Schaa, and D. Kaeli, " Multi2Sim: A Simulation Framework for CPU-GPU Computing ," in *PACT*, Sep. 2012.

[23] " AMD Accelerated Parallel Processing (APP) Software Development Kit ," 2016. [Online]. Available: http://developer.amd.com/sdks/amdappsdk/

[24] C. Silvano, G. Palermo, S. Xydis, and I. S. Stamelakos, "Voltage island management in near threshold manycore architectures to mitigate dark silicon," in *DATE, Dresden, Germany, March 24-28, 2014*, 2014, pp. 1–6.

[25] M. Seok, G. Chen, S. Hanson, M. Wieckowski, D. Blaaw, and D. Sylvester, "Cas-fest 2010: Mitigating variability in near-threshold computing," in *J. Emerg Selec. Topics Cir. Sys*, vol. 1, no. 1, 2011, pp. 42–49.

DocuSign Envelope ID: 318023CC-46D7-4A3D-868B-AD82E9E540A5

ABSTRACT

A Study of Distributions and Travel Paths to Minimize Navigation Uncertainty Using Line of Sight Measurements to Known Landmarks in GPS-Denied Environments

by

Ryan D. Lamoreaux

Utah State University, 2016

Major Professor: Rajnikant Sharma, PhD
Department: Electrical and Computer Engineering

To date, Global Positioning Systems (GPS) occupy a crucial place in most navigation systems. Because reliable GPS is not available everywhere, navigation within GPS-denied environments is an area of deep interest in both military and civilian applications. Much work has been done on this topic and many have explored vision-aided inertial navigation as a possible solution. One form of vision-aided navigation measures the direction from the vehicle and a feature or landmark of known location to deduct information about the vehicles position and attitude. This work will attempt to provide insight into two questions related to this type of navigation. First, given a set travel path and the freedom to place landmarks of known locations in any distribution, what is the minimum number of landmarks required to achieve a desired navigation accuracy as a vehicle travels from point to point? Second, given a set distribution of known landmarks, what is the shortest path a vehicle may travel from point to point to achieve a desired navigation accuracy? This work will attempt to answer these questions by incorporating line of sight measurements (LOS) into a modeled small unmanned aerial vehicle. This system will be validated through the use of Monte Carlo simulations. Cost functions will be developed to aid in the selection of an optimal trajectory and an optimal landmark distributions to compare. Monte Carlo simulations will

DocuSign Envelope ID: 318023CC-46D7-4A3D-868B-AD82E9E540A5

be run on each proposed solution to determine validity by showing whether the uncertainty remains within requirements or not.

(19 pages)